

Spectral analysis II

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Please open Python notebook [03_spectral_2](#)

Agenda

- **DFT refresher**
- DFT of typical signals
- Properties of DFT
- Some more math on DFT
- Spectrogram and what can we do with it
- Summary and “take-home” messages

Discrete Fourier transform (DFT) analysis

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi \frac{k}{N} n}$$

- Input: signal $x[n]$ of N samples
- Output: N **complex** coefficients $X[k]$ providing the information on
 - Frequency – index k , corresponds to normalized frequency k/N and regular frequency $k/N F_s$. Normalization and de-normalization of frequency involves a simple division or multiplication by sampling frequency F_s .
 - How much ? Magnitude (absolute value, modul) $|X[k]|$.
 - How shifted ? Phase (angle, argument, phase shift) $\arg X[k]$.
- Easy to implement by definition but using FFT is much faster (for $N = 2^b$)

DFT analysis II.

- For real input signals $x[n]$, coefficients $X[k]$ are symmetrical (complex conjugate $X[k] = X^*[N-k]$):
 - $|X[k]| = |X[N-k]|$ and $\arg X[k] = -\arg X[N-k]$
- ... so that it is enough to store and visualize the spectrum only from $k = 0$ to $k = N/2$. This corresponds to normalized frequencies $0 \dots \frac{1}{2}$ and regular frequencies $0 \dots F_s/2$.
- Attention, the number of coefficients to retain is **NOT** $N/2$ but $N/2+1$!!!

#dft_anal

DFT synthesis - IDFT

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{+j2\pi \frac{k}{N} n}$$

- Input: N **complex** coefficients $X[k]$
- Output: N samples of signal $x_s[n]$.
 - In case we did nothing with the spectrum, $x_s[n]$ is exactly the same as $x[n]$.
- Synthesis by DFT involves summing pairs of complex exponentials at k and $N-k$ running “against each other”, creating one shifted cosine – see derivation in the last lecture.
 - When computing, DFT/FFT can produce very small imaginary components, better to kill them using `np.real()`.

#dft_synt

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DFT of a D.C. (stejnoseměrný) / constant signal

- $x[n] = a$

$$X[k] = \sum_{n=0}^{N-1} a e^{-j2\pi \frac{k}{N} n}$$

- For $k=0$, the sum is $X[0] = Na$.
- For everything else, the sum of k “revolutions” of the complex exponential is zero.
- Intuition check: D.C. has just zero frequency, everything else should be zero - **OK**

Very slow signal $\Leftrightarrow$ narrow spectrum

#dft_of_constant

DFT of a pulse (**impuls**) at time zero

- $x[0] = a$, otherwise $x[n] = 0$.

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi \frac{k}{N} n} = a e^{-j2\pi \frac{k}{N} 0} = a e^0 = a$$

- Only the term for $n=0$ will produce something here, the complex exponential is $\exp(0) = 1$, so $X[k] = a$ for all k 's
- Intuition check: the pulse is very fast (like a click), should contains lots of frequencies - **OK**

Very fast signal $\Leftrightarrow$ wide spectrum

#dft_of_pulse

DFT of a shifted pulse (posunutý impuls)

- $x[g] = a$, otherwise $x[n] = 0$. g is the position of the pulse.

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi \frac{k}{N} n} = a e^{-j2\pi \frac{k}{N} g}$$

- Only the term for $n=g$ will produce something - here, the complex exponential is $\exp(-2\pi g k / N)$, so
 - $X[k] = a \exp(-2\pi g k / N)$.
 - Magnitude is $|X[k]| = a$
 - Phase is $\arg(a \exp(-2\pi g k / N)) = -2\pi g k / N$... linear with slope $-2\pi g / N$
- For visualization, you might try using `np.unwrap` to see values of phase outside of $-\pi \dots +\pi$.
- Intuition check: the signal only shifted, the magnitudes should not change, shift should be reflected in phases - **OK**

#dft_of_shifted_pulse

DFT of a rectangular pulse (pravoúhlý / obdélníkový impuls)

- $x[n] = a$ for $n = 0 \dots g-1$, zero otherwise (so it will have g non-zero samples).
- No derivation (we'll see this later) but remember that wide signal should have narrow spectrum and narrow signal should have wide spectrum ...

#dft_of_rectangle

- The function we see is $\sin(x) / x$, so called **cardinal sine** (kardinální sinus). We'll be seeing this a lot ...
- For narrow rectangles, the cardinal sine lobes (laloky) seem to be distorted as we move towards $F_s/2$ – this is caused by **aliasing** (more about it in the lecture on sampling)

DFT of a “symmetrical” rectangular pulse

- Trying to make the rectangle symmetrical so that the phase is not shifted.
- Will work only for odd (**liche**) g by setting the same amount of samples for negative n as for positive n . For example, for $g = 5$, set $x[-2 \dots +2] = a$
- However, we can not work with negative indices, so we need to consider the end of input buffer as negative indices ...
 $x[0 \dots 2] = a, x[N-2] = a, x[N-1] = a$

#dft_of_symmetrical_rectangle

- The phase is either 0 or $\pm\pi$, this means that this spectrum is real – just positive or negative numbers !

Symmetrical signal $\Leftrightarrow$ real spectrum

DFT of cosine with the period of N samples

$$x[n] = A \cos\left(2\pi \frac{1}{N}n + \phi\right)$$

- We could run the DFT analysis, but let's work the "lazy" way:

- Split the cosine using our well known formula $\cos \alpha = \frac{e^{j\alpha} + e^{-j\alpha}}{2}$

$$x[n] = \frac{A}{2} e^{j\phi} e^{j2\pi \frac{1}{N}n} + \frac{A}{2} e^{-j\phi} e^{-j2\pi \frac{1}{N}n}$$

- Write the synthesis formula next to it and try to see if we can find respective coefficients

$$x_s[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{+j2\pi \frac{k}{N}n}$$

- We easily find $X[1] = N \frac{A}{2} e^{j\phi}$

- The next one we could find is $X[-1]$ which is not there (can use only indices from 0 ... $N-1$) but we already know about the $X[-1] = X[N-1]$: $X[N-1] = N \frac{A}{2} e^{-j\phi}$

- Coefficients $X[1]$ and $X[N-1]$ for a cosine are almost the same, except for the $|X[1]| = |X[N-1]| = N \frac{A}{2}$ $\arg X[1] = \phi$, $\arg X[N-1] = -\phi$ $X[1] = X^*[N-1]$

#dft_of_cos1 - OMG, what's going on with the phase ???

DFT of cosine with more periods in N samples

$$x[n] = A \cos(2\pi \frac{g}{N}n + \phi)$$

- We could find the values of coefficients exactly in the same way, just looking at index g instead of 1
- Exactly the same, only the index will change $X[g] = N \frac{A}{2} e^{j\phi}$

$$X[-g] = X[N - g] \quad X[N - g] = N \frac{A}{2} e^{-j\phi}$$

- Magnitudes are the same, arguments opposite:

$$|X[g]| = |X[N - g]| = N \frac{A}{2} \quad \arg X[g] = \phi, \quad \arg X[N - g] = -\phi \quad X[g] = X^*[N - g]$$

#dft_of_cos_multiperiod

DFT of a cosine that does not fit exactly N samples ...

- In real life, we can not say “I’m going to analyze only signals with period of 128 samples”. We obtain an input and we have to go ...
- So it can happen (and it regularly happens) that our cos is “badly cut”.
- Remember: any sharp edge in a signal generates high frequencies.

#dft_of_general_cos

For a period that does not fit into N exactly k times (“wrong cut” of cos), the spectrum is “leaking” to neighboring frequencies.

Why ? Because we're windowing our signal ...

- $x[n] = w[n] s[n]$, where $w[n]$ is the window function.
- In case of multiplication of signals in time, we'll have a **convolution** (konvolute) of spectra in frequency.
- More about convolution in the filtering lecture, for cosine signal, it simply means, that the spectrum of $w[n]$ shifts to its frequency
- So what is the spectrum of our window ?
 - Analyzing N samples is not enough, we'd get a constant (and we know it's spectrum is a single coefficient).
 - Increasing N to more samples by zero padding: $256 \rightarrow N_{fft} = 4096$ (16x)
 - Zoom only on low frequencies to see details ... and showing only magnitude
 - ... also showing where are the points of length N DFT !

#dft_rectangle_zeropad

Better frequency resolution for a “good” cosine

$$x[n] = A \cos(2\pi \frac{g}{N} n + \phi)$$

#dft_of_cos_multiperiod_detail

- Here, we see the cardinal sine, but the frequencies of the original DFT sample it exactly at zero values !

Better frequency resolution for a “bad” cosine

#dft_of_cos_generalcos_detail

- We see the cardinal sine, and we are not lucky, so the theoretical 1 line “leaks” into neighboring frequencies.
- We can try to fix it by another window ... Hamming

$$w[n] = 0.54 - 0.46 \cos \frac{2\pi n}{N-1} \quad \text{for } 0 \leq n \leq N-1, \quad 0 \text{ elsewhere}$$

#dft_of_generalcos_hamming

- Less “messy” spectrum than the rectangular window (the edges resulting from “bad cutting” are attenuated) but also worse frequency resolution.
- Remember: smoother signal $\Leftrightarrow$ wider spectrum.

DFT of something periodic

- Periodic rectangular pulses with period N_{pulse}
- Fitting N samples with an integer number of periods

#dft_pulsetrain

- Compared to the spectrum of one rectangular pulse, the spectrum is **sampled**.
- The positions of samples correspond to $1/N_{pulse}$ (normalized frequency) and F_s / N_{pulse} (regular frequency) – more in a separate lecture on periodic signals.

Periodic signal $\Leftrightarrow$ sampled “discrete” spectrum

DFT of a periodic signal with a “wild” period

- Periodic rectangular pulses with period N_{pulse}
- Not fitting N samples with an integer number of periods (which is the usual situation ...)

#dft_pulsetrain_general

- It does something but there is a brutal “leaking” into neighboring frequencies impacting also the shape of the spectrum.
- However, this is what we’re doing most of the time – see the speech signal #dft_anal at the beginning of this lecture

In case we know the period, or are able to measure it precisely, DTFT might be a safer option.

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Scaling

- If the signal is multiplied with a constant, $y[n] = a x[n]$, the DFT coefficients will be $Y[k] = a X[k]$
- Proof:

$$X[k] = \sum_{n=0}^{N-1} ax[n]e^{-j2\pi \frac{k}{N}n} = a \sum_{n=0}^{N-1} x[n]e^{-j2\pi \frac{k}{N}n} = aX[k]$$

#dft_scaling

Additivity

- If
 - DFT for signal $x_1[n]$ is $X_1[k]$
 - and DFT for signal $x_2[n]$ is $X_2[k]$
- Then, if we sum the signals: $y[n] = x_1[n] + x_2[n]$
- The resulting DFT will be also the sum of the original ones:
 $Y[k] = X_1[k] + X_2[k]$
- Proof:

#dft_additivity

... why

$$X[k] = \sum_{n=0}^{N-1} (x_1[n] + x_2[n])e^{-j2\pi \frac{k}{N}n} = \sum_{n=0}^{N-1} x_1[n]e^{-j2\pi \frac{k}{N}n} + \sum_{n=0}^{N-1} x_2[n]e^{-j2\pi \frac{k}{N}n} = X_1[k] + X_2[k]$$

Linearity

- Scaling and additivity together
- If DFT for signal $x_1[n]$ is $X_1[k]$ and DFT for signal $x_2[n]$ is $X_2[k]$
- Then, if we do a **weighted sum** the signals:
 $y[n] = a_1 x_1[n] + a_2 x_2[n]$
- the resulting spectrum will be exactly the same weighted combination of the original ones: $Y[k] = a_1 X_1[k] + a_2 X_2[k]$

$$X[k] = \sum_{n=0}^{N-1} (a_1 x_1[n] + a_2 x_2[n]) e^{-j2\pi \frac{k}{N} n} = a_1 \sum_{n=0}^{N-1} x_1[n] e^{-j2\pi \frac{k}{N} n} + a_2 \sum_{n=0}^{N-1} x_2[n] e^{-j2\pi \frac{k}{N} n} = a_1 X_1[k] + a_2 X_2[k]$$

#dft_linearity

Linearity is absolutely crucial in many applications, and we intentionally use non-linearities in others – neural networks.



Shifts of short signals

- Consider a relatively short signal $x[n]$ that has DFT $X[k]$.
- What happens if this signal is shifted to the right (delayed) by m samples: $y[n] = x[n-m]$ (let's consider that the signal does not leave the interval $0 \dots N-1$)
- The complex exponentials “seeing” the signal will be m samples “older”, so that we can re-write the DFT as

$$\begin{aligned} Y[k] &= \sum_{n=0}^{N-1} y[n] e^{-j2\pi \frac{k}{N} n} = \sum_{n=0}^{N-1} x[n-m] e^{-j2\pi \frac{k}{N} n} = \sum_{n=0}^{N-1} x[n] e^{-j2\pi \frac{k}{N} (n+m)} = \sum_{n=0}^{N-1} x[n] e^{-j2\pi \frac{k}{N} (n+m)} = \\ &= \sum_{n=0}^{N-1} x[n] e^{-j2\pi \frac{k}{N} n} e^{-j2\pi \frac{k}{N} m} = e^{-j2\pi \frac{k}{N} m} \sum_{n=0}^{N-1} x[n] e^{-j2\pi \frac{k}{N} n} = e^{-j2\pi \frac{k}{N} m} X[k] \end{aligned}$$

- The resulting coefficient is the original one multiplied with a “correction factor”:

$$Y[k] = e^{-j2\pi \frac{k}{N}m} X[k]$$

- The magnitude of the correction is 1, so
- The phase of the correction is shifted by : $|Y[k]| = |e^{-j2\pi \frac{k}{N}m}| |X[k]| = |X[k]|$

$$\arg Y[k] = \arg e^{-j2\pi \frac{k}{N}m} + \arg X[k] = \arg X[k] - 2\pi \frac{k}{N}m$$

The magnitude spectrum will be the same, the phase one will be inclined “downhill” for delay and “uphill” for advancing the signal.

#dft_shifted_lin

Shift of a signal that covers the whole period

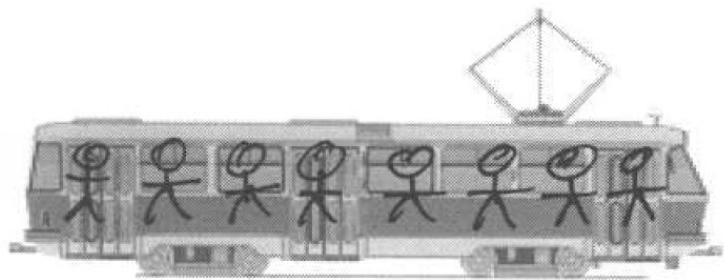
- Consider a longer signal, for example a cosine nicely fitting in N samples
- When shifted, “gaps” appear, and creates edges generating unwanted frequencies ...

#dft_shifted_lin_long

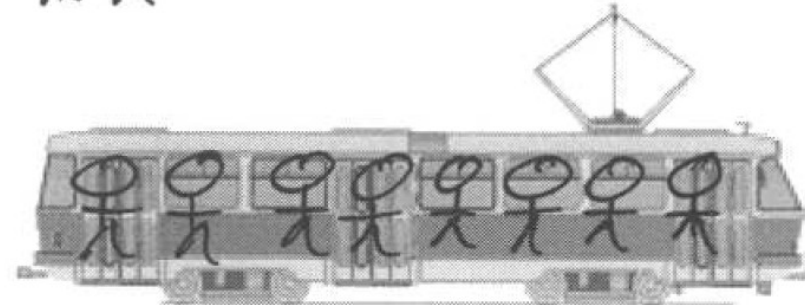
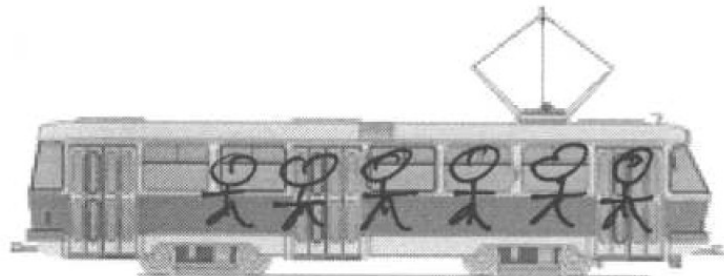
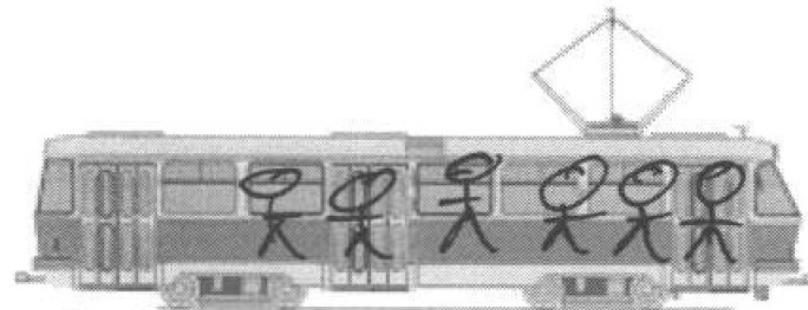
Ooops, that's a mess 😞

Correct way to shift signals for DFT ...

- Remember: we need to always stay in the buffer $0 \dots N-1$
- In case m samples of the signal are pushed out at the end, they need to return to the beginning.



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0 2 místa



Circular shift mathematically

- Modulo N function always returns values from $0 \dots N-1$
- Implementing circular shift by modulo indexing:
 $y[n] = x[\text{mod}_N(n-m)]$
- For such a circular shift, we can safely use the correction computed above - the resulting coefficient is the original one multiplied with a "correction factor":
 - The magnitude of the correction is 1 so
 - The phase of the correction is $-2\pi \frac{k}{N} m$

$$Y[k] = e^{-j2\pi \frac{k}{N} m} X[k]$$

$$|Y[k]| = |e^{-j2\pi \frac{k}{N} m}| |X[k]| = |X[k]|$$

The magnitude spectrum will be the same, the phase one will be inclined "downwards" by $-2\pi \frac{k}{N} m$, meaning the signal is time shifted.

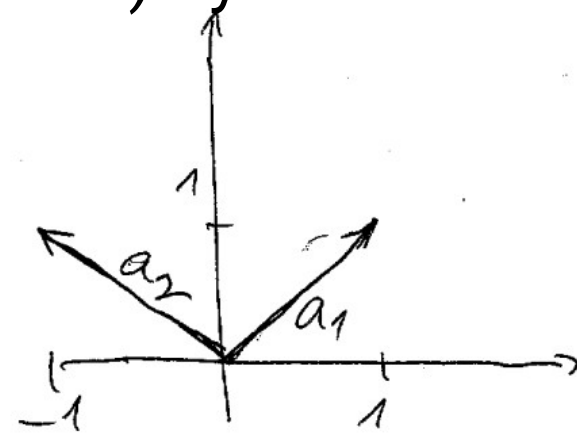
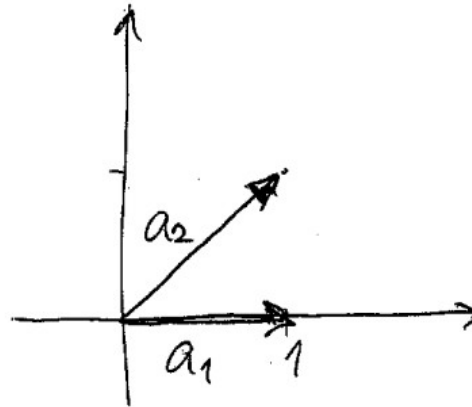
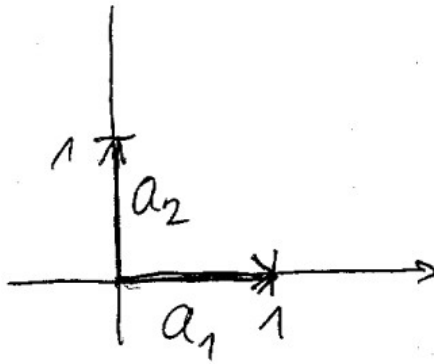
#dft_shift $\arg Y[k] = \arg e^{-j2\pi \frac{k}{N} m} + \arg X[k] = \arg X[k] - 2\pi \frac{k}{N} m$

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Orthogonality

- For 2D, we can check the orthogonality ([kolmost](#)) by hand ...



- Mathematically, checking by dot-product
 - $[1 \ 0] [0 \ 1]^T = 1 \times 0 + 0 \times 1 = 0$ **orthogonal**
 - $[1 \ 0] [1 \ 1]^T = 1 \times 0 + 1 \times 1 = 1$ **not orthogonal**
 - $[1 \ 1] [-1 \ 1]^T = 1 \times (-1) + 1 \times 1 = 0$ **orthogonal**

Are DFT bases orthogonal ?

- We'll check it by a standard dot-product over N samples.

$$c = \sum_{n=0}^{N-1} a_k[n] a_l[n] = 0, \quad \text{for } k \neq l$$

- But we have complex bases, so need to do complex conjugation (as for projection)

- For DFT bases, we $c = \sum_{n=0}^{N-1} a_k[n] a_l^*[n] = 0, \quad \text{for } k \neq l$

$$c = \sum_{n=0}^{N-1} e^{-j2\pi \frac{k}{N} n} e^{-(-j2\pi \frac{k}{N} l)} = \sum_{n=0}^{N-1} e^{j2\pi \frac{l-k}{N} n} = 0, \quad \text{for } k \neq l$$

Are DFT bases normal ?

- Norm of each basis must be 1 ...

- For DFT:
$$\|a[n]\|_2 = \sum_{n=0}^{N-1} |a[n]|^2$$

- The norm is not 1, $\|a[n]\|_2 = \sum_{n=0}^{N-1} |e^{+j2\pi \frac{k}{N} n}|^2 = \sum_{n=0}^{N-1} 1 = N$ IDFT when synthesizing the signal:

- Remember the multiplication by N when synthesizing the signal for a cosine !

After IDFT fix, DFT bases are normal,
$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{+j2\pi \frac{k}{N} n}$$
 normal.

$$|X[g]| = |X[N - g]| = N \frac{A}{2}$$

DFT extending beyond $k=0 \dots N-1$ – periodicity

- We have defined DFT to strictly compute the spectrum for $k=0 \dots N-1$, this corresponds to
 - Normalized frequencies from 0 to almost 1 (precisely $(N-1) / N$)
 - Regular frequencies from 0 to almost F_s (precisely $F_s(N-1) / N$)
 - And we usually visualize it only till $N/2$ (i.e. till $\frac{1}{2}$ in normalized frequencies and $F_s/2$ in regular ones).
- What happens if we go beyond interval $k=0 \dots N-1$?
- Augmenting k by a multiple of the number of samples N :

$$X[k + gN] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi \frac{k+gN}{N} n} = \sum_{n=0}^{N-1} x[n] e^{-j2\pi (\frac{k}{N} + \frac{gN}{N}) n} = \sum_{n=0}^{N-1} x[n] e^{-j2\pi \frac{k}{N} n} e^{-j2\pi \frac{gN}{N} n} =$$

$$\sum_{n=0}^{N-1} x[n] e^{-j2\pi \frac{k}{N} n} e^{-j2\pi g n} = \sum_{n=0}^{N-1} x[n] e^{-j2\pi \frac{k}{N} n} 1 = X[k]$$

- DFT is periodic with N samples. In frequencies, this corresponds to periodicity
 - With period of 1 (normalized frequencies)
 - With period of F_s (regular frequencies)
- This is the “mathematical” proof of periodicity, we will see another one in the lecture on sampling !
- We can do a similar proof for DTFT

#dft_periodicity

$$\tilde{X}(e^{j(\omega+g2\pi)}) = \sum_{n=0}^{N-1} x[n]e^{-j(\omega+g2\pi)n} = \sum_{n=0}^{N-1} x[n]e^{-j(\omega n+g2\pi n)} =$$

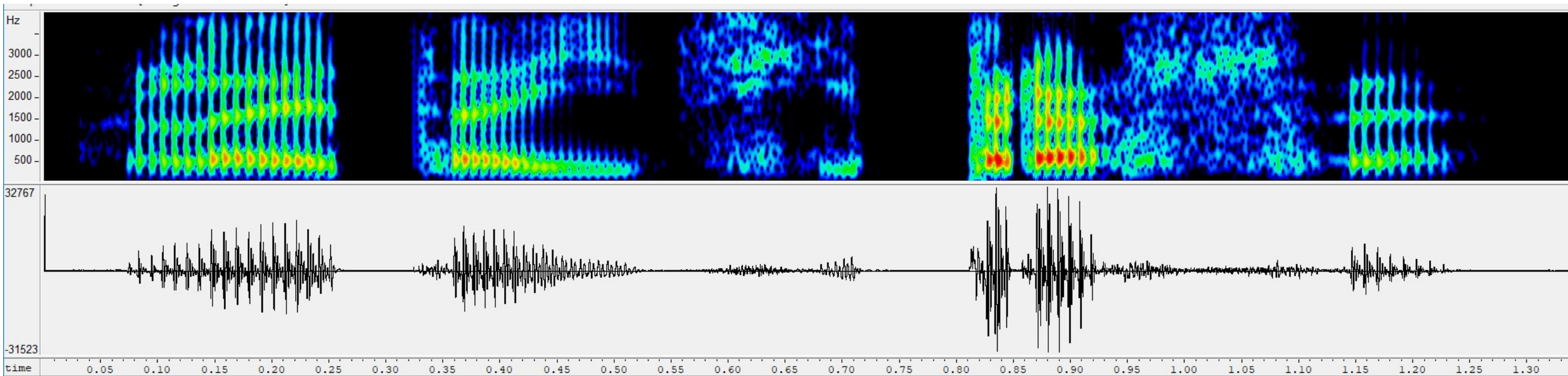
$$\sum_{n=0}^{N-1} x[n]e^{-j\omega n}e^{-jgn2\pi} = \sum_{n=0}^{N-1} x[n]e^{-j\omega n}1 = \tilde{X}(e^{j\omega})$$

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Spectrogram

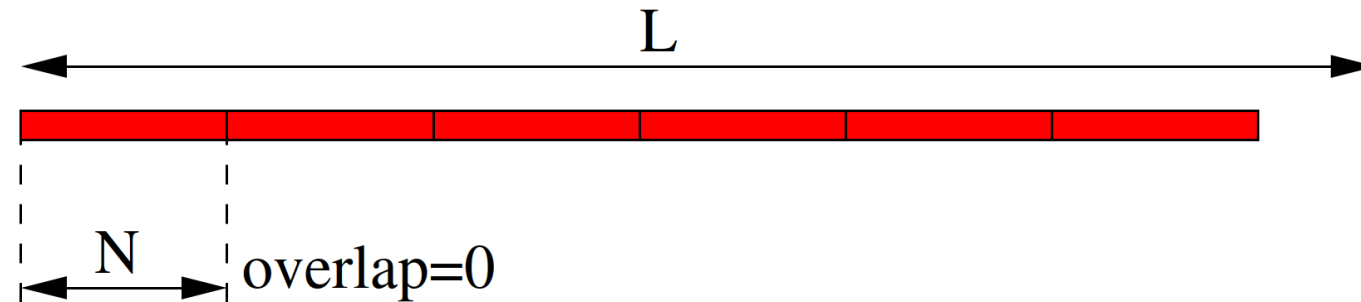
- We want to see the evolution of spectrum over time.
- For our speech signal, want something like this:



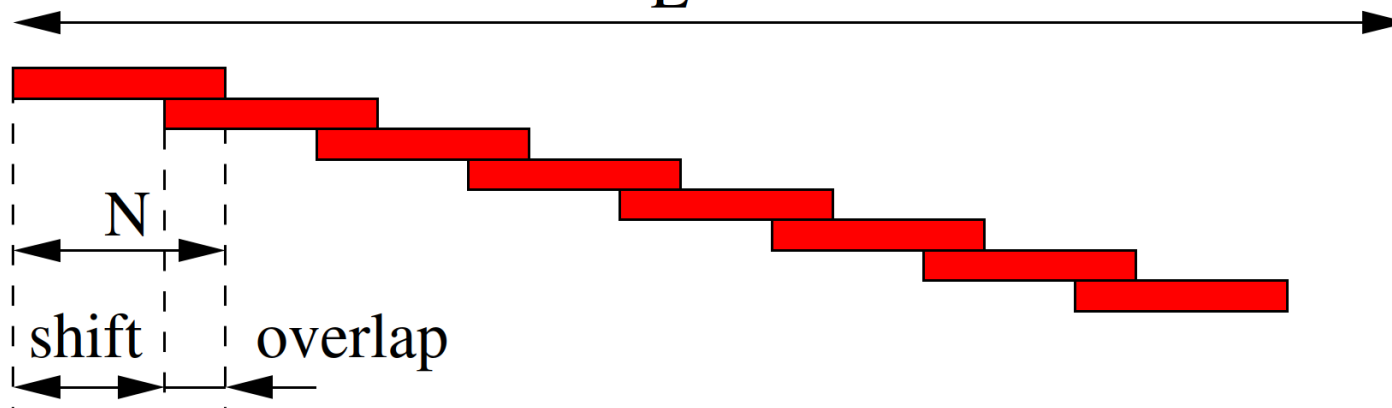
- Calling `#spectrogram_blackbox` without knowing what it really does.
- We better learn **what the spectrogram does exactly**

Pre-processing of the signal before computation

- Division of a long signal into short segments – frames (rámce)
- Segmentation without overlap



- Segmentation with overlap (more usual)



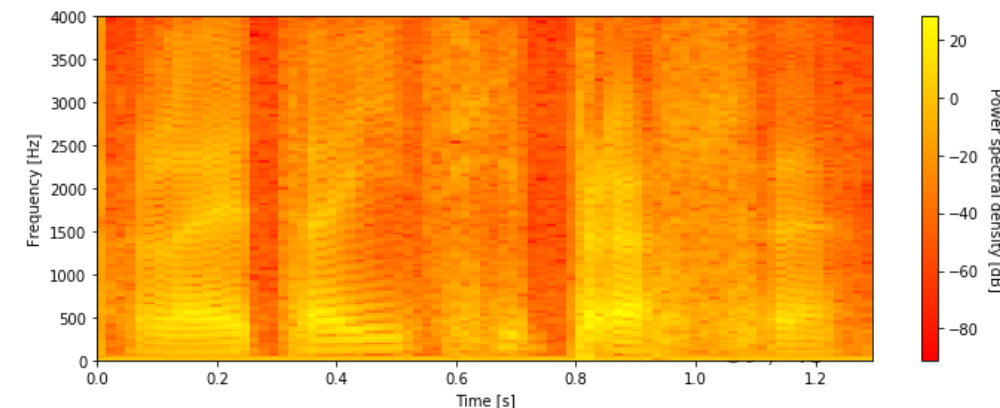
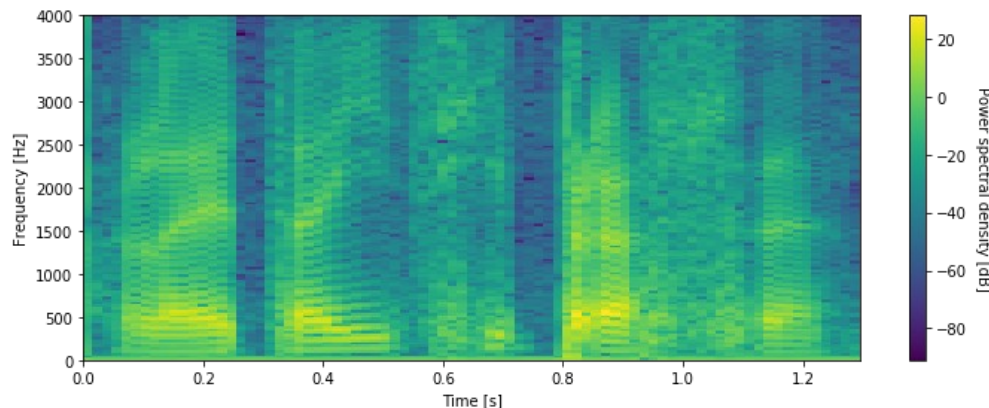
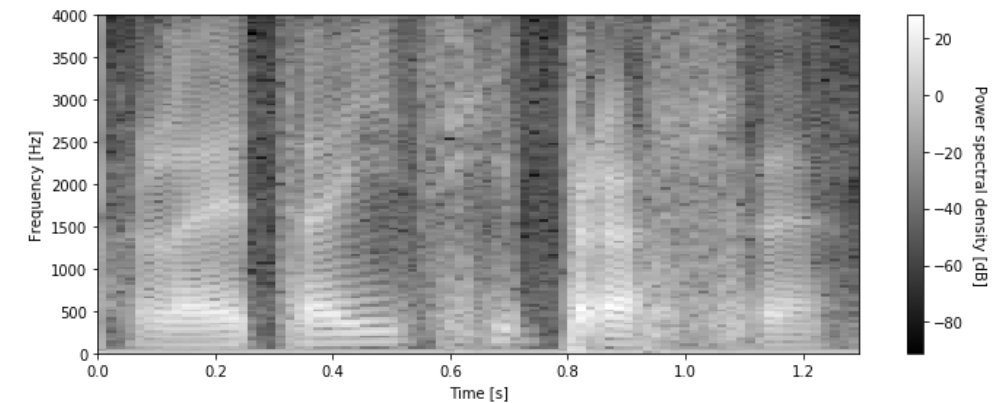
Windowing the frames and running DFT for all ...

- Window each frame by a selected window
 - Rectangular – good frequency selectivity, more mess at higher frequencies.
 - Hamming – worse frequency selectivity, less mess at higher frequencies.
 - Hanning (or Hann) window – a sequence overlapping by $N/2$ sums up to 1 – see later. Beware of confusion 'mm' vs 'nn' ...
- If needed, zero-pad to reach the desired number of point N_{fft} .
- As usual, select only $0... N/2$, resp $0... N_{fft}/2$ points.
- Visualize

#spectrogram

More on spectrogram visualization

- We usually take the log of so called power spectral density – square of spectrum $10 \log_{10} |X[k]|^2$ and claim it is in decibel [dB]
- Selection of colormap ...
- Playing with brightness, contrast, etc ...



Short-term vs long-term spectrogram

- Wanted: more details in frequency => set longer frame length => however, less precision in time (averaging details over longer time...) – **long-term spectrogram**
- Wanted: more details in time => set shorter frame length (eventually with more FFT points to have a nice figure) => however, less precision in frequency (the frame “sees” less signal) – **short-term spectrogram**
- Wanted: more frames per second => set more overlap (less frame shift) => however, will not be able to detect short events (still averaged by the frame length...

#spectrogram_long_short_term

Setting good spectrogram parameters is a tricky business and there is no universal solution – ask colleagues and then tune the parameters.

Synthesizing signal from spectrogram

- Need to be careful with the analysis – define frames with $\frac{1}{2}$ frame overlap and use Hanning (Hann) window – their sequence sums exactly to 1.
- When synthesizing, initialize a buffer with the length of the resulting signal by zeros, then run a cycle:
 - For each frame, take spectrum and complete the upper (from $N/2+1$... $N-1$) by the symmetry $X[k] = X^*[N-k]$
 - Perform IDFT that generates the current frame.
 - Add it to the buffer
 - Move to the next frame, shift the position where signal will be added to the buffer by frame overlap.

#dft_synt

Playing with spectrogram I – filtering

- Without changes to the spectrogram, the Hanning window can be applied either in the analysis (before DFT) or in the synthesis (after IDFT)
- However, if spectrogram is modified, it is good to perform the windowing on both ends. However, the whole thing needs to be still Hanning window, so:
 - Apply sqrt of Hanning before DFT
 - Apply sqrt of Hanning after IDFT
 - $\text{sqrt Hanning} * \text{sqrt Hanning} = \text{Hanning}$.

#spectrogram_filtering – deleting part of spectrogram + up to you to try anything else

Playing with spectrogram II – duration modification

- Taking every 2nd frame from the spectrum – shortening 2x – “Ostrava accent”
- Repeating every frame – lengthening 2x – “Prague accent”

#spectrogram_duration_modif

The quality of audio is not perfect – the phases would need better processing (continuity of phase across frames).

Agenda

- DFT refresher
- DFT of typical signals
- Properties of DFT
- Some more math on DFT
- Spectrogram and what can we do with it
- **Summary and “take-home” messages**

SUMMARY

- DFT (computed by FFT) is the basic tool for frequency analysis.
- Basic truths on signals and spectra:
 - Narrow signal $\Leftrightarrow$ wide spectrum
 - Wide (smooth) signal $\Leftrightarrow$ narrow spectrum
 - Periodic signal $\Leftrightarrow$ discrete spectrum
 - Discrete signal $\Leftrightarrow$ periodic spectrum
- Shifts of signals
 - Delay $\Rightarrow$ magnitude spectrum stays the same, phase goes downhill.
 - Advance $\Rightarrow$ magnitude spectrum stays the same, phase goes uphill.
 - For DFT, we need to do circular shifts to stay in $0 \dots N-1$

SUMMARY II.

- Spectrogram is evolution of spectrum in time
 - Contains magnitude and phase, only magnitude is shown
 - Lots of tuning can be done by non-linearity (log), colormaps, and setting brightness and contrast of the image.
 - Better frequency or better time resolution, but not both – long vs. short-term spectrogram
- Synthesis of signal from spectrogram
 - Analysis should be done carefully – shift is $\frac{1}{2}$ of the frame-length, Hanning window.
 - Spectrum needs to be completed by the 2nd half before IDFT.
 - In case modifications are done, it's safer to use sequence of $\sqrt{\text{Hanning}}$ and $\sqrt{\text{Hanning}}$
- Filtering is usually done by other means, but spectrogram modifications are of great use in machine learning (learned T-F masks, etc).